Milestone E: Chart Integration - Delivery Report

Date: 31 January 2026

Milestone: Milestone E - Chart Integration

Status: ✓ Completed

Preview URL: science-canvas-creator.vercel.app

Executive Summary

Milestone E completed successfully. A chart creation and editing system has been integrated into the canvas editor for publication-quality charts. The implementation uses an immediate-insert workflow with Properties panel configuration (rather than a pre-insert dialog), improving UX and workflow consistency. All deliverables implemented, all tests passing.

Deliverables Completed

Chart Creation Workflow

1. Click chart tool icon in left toolbar
2. Click canvas to insert default chart (bar chart with sample data)
3. Configure via Properties panel
4. Edit data through data table dialog

Provides immediate visual feedback and matches other canvas tool patterns.

Chart Data Management

- Editable spreadsheet-style table
- Click headers to rename columns
- Edit cells directly (focus selects all, empty cells default to "0")
- Add/remove rows and columns
- Changes reflect immediately in chart preview

Chart Types Supported

- **Bar Chart** - Vertical bars with grouped series support
- **Line Chart** - Connected data points with multiple series
- **Scatter Plot** - Individual data points for correlation visualization
- **Box Plot** - Statistical distribution with median, quartiles, and outliers

Column Mapping Configuration

- **X-Axis Column:** Category labels or independent variable data
- **Y-Axis Columns:** One or more columns for data series (enables grouped/multi-series charts)
- Charts auto-update when assignments change

Error Bars System

- Enable via checkbox in Chart Type section
- Assign error columns individually per series
- Configure appearance per series (match series color or set custom color)
- Auto-scale and position based on data values
- Not available for box plots (inherently show statistical variation)

Chart Styling & Customization

Typography:

- Font family selector (Google Fonts and system fonts)
- Individual font sizes for chart title, axis labels (both axes), and legend

Labels & Text:

- Editable chart title, X-axis and Y-axis labels
- Labels default to column names

Visual:

- Series color pickers per data series
- Background color selection and transparent background toggle
- Show/hide legend toggle

Implementation Differences from Original Specification

Workflow:

- Original: Single dialog with data entry, type selection, and customization before insertion
- Delivered: Immediate insertion, then configure via Properties panel with separate data dialog

Error Bars:

- Original: Single error column with type selector (SD/SEM)
- Delivered: Per-series error column assignment with color customization

Benefits:

- Faster workflow (immediate chart visibility)
- Consistent with other tool patterns
- More flexible error bars for multi-series charts
- Better visual feedback

Testing Results

All tests updated and passing:

Automated: Chart creation/insertion, data table editing, type switching, column mapping, error bar configuration, style customization, updates/re-rendering

Manual: All four chart types render correctly, multi-series charts with legends, error bar display/scaling, color customization, font controls, transparent backgrounds, canvas resizing without quality loss, concurrent update prevention

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Additional Work Completed Outside Milestone Scope

Database IO Performance Optimization

Context: Analysis of `dev_io_log` revealed 2.2 GB data transfer bottleneck.

Investigation: Supabase CLI analysis identified three bottlenecks in `project_versions`, `ai_icon_generations`, and `canvas_projects` tables. Root cause: `SELECT *` queries fetching multi-MB JSON/BLOB columns unnecessarily.

Optimizations:

1. **Version Management** (`src/lib/versionManager.ts`)
    - excluded canvas_data when not required (5.95 MB → 5 KB, 99.9% reduction)
    - Optimized functions to use metadata-only when appropriate (70 MB → 1 KB, 99.99% reduction)
    - Applied a 50-version default limit
2. **Community Gallery** (`src/pages/Community.tsx`)
    - Excluded canvas_data (1.19 MB → 15 KB, 99% reduction)
3. **Featured Projects**
    - Optimized queries to exclude unnecessary data

Results:

- Overall IO: **2.2 GB → ~50 MB (98% reduction)**
- Version history load: **2-23s → <0.5s**
- Gallery load: **2s → <0.3s**
- No UX changes

Conclusion

Milestone E delivered in full:

- ✓ Four chart types for scientific visualization
- ✓ Flexible data entry/editing
- ✓ Column mapping for single/multi-series charts
- ✓ Advanced error bar configuration (per-series customization)
- ✓ Comprehensive styling controls
- ✓ Seamless canvas integration
- ✓ Live updates without duplication

Additionally:

- ✓ database performance optimizations achieved 98% IO reduction

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